DB PENSION PLAN FUNDING SIMULATOR: ANALYSIS

ABSTRACT

This funding calculator gives summary DB plan returns statistics and uses bootstrapped historical returns from 2013 to 2025 to generate 10-year horizon distributions for the plan funding ratio and surplus/deficit.

Asset portfolio pre-set weightings from a typical 2011 or 2025 plan can be selected, or weights entered manually. Asset allocations are drawn from the PPF Purple Books for 2011 and 2025 with ambiguous asset classes or those lacking data ignored or imputed using independence assumptions.

Liabilities are modelled as a mix of nominal and inflation-linked bonds with approximate maturity weightings matching that of the demographics of schemes open, closed to new members and closed to new benefit accrual in 2025.

ETF proxies with Geltner (1991) smoothing/unsmoothing are used for private asset classes. MA/ARMA models developed in Getmansky-Lo-Makarov (2003) were not used due to their private asset subclass specificities.

ANALYSIS

If DB plan assets and liabilities are assumed to follow normally distributed returns, then changes in FR (≈ change in assets – changes in liabilities) would be normally distributed and so FR would be lognormal. The bulk of 2011 and 2025 10Y FR horizon distributions appear lognormal, but changes in FR are platykurtic (kurtosis ≈ 1.4 < 3) for the 2011 mature allocation and leptokurtic (kurtosis ≈ 4.0 > 3) for the 2025 mature allocation (see Figures 1 to 4).

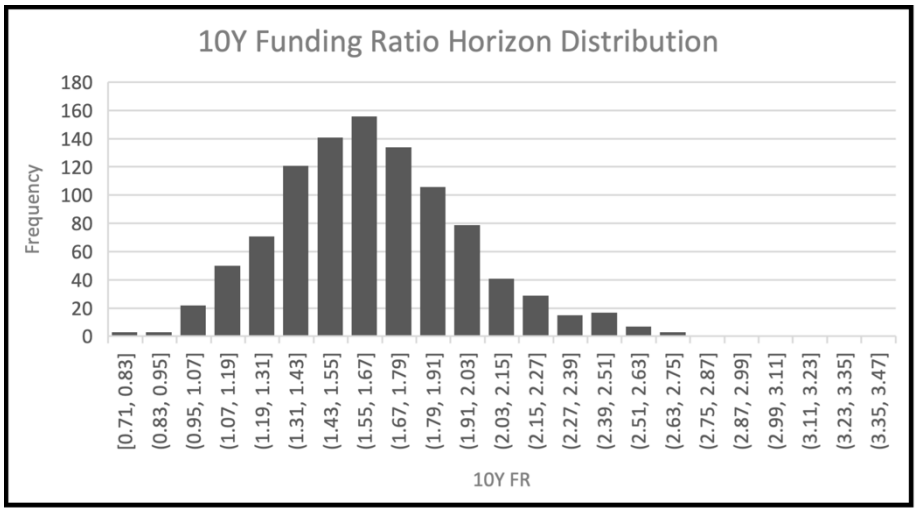

Figure 1: 10Y FR horizon distribution for 2011 mature asset allocation

MONTHLY PORTFOLIO STATISTICS					
Beginning Plan Assets (£m):	150.6	***FR Mean Change:***	**0.4213**	***FS Mean Change (£m):***	**63.4**
Beginning Plan Liabilities (£m):	150.8	***FR SD:***	**1.8699**	***FS 100P% VaR (£m):***	**-411.4**
Confidence Level: P =	0.95	***FR Skew:***	**-0.5037**	***FS 100P% CVaR (£m):***	**-598.8**
Beginning Funding Ratio:	**1.00**	***FR Kurtosis:***	**1.3763**	***FS Max Drawdown (£m):***	**-1040.7**

Figure 2: 2011 mature asset allocation monthly portfolio statistics

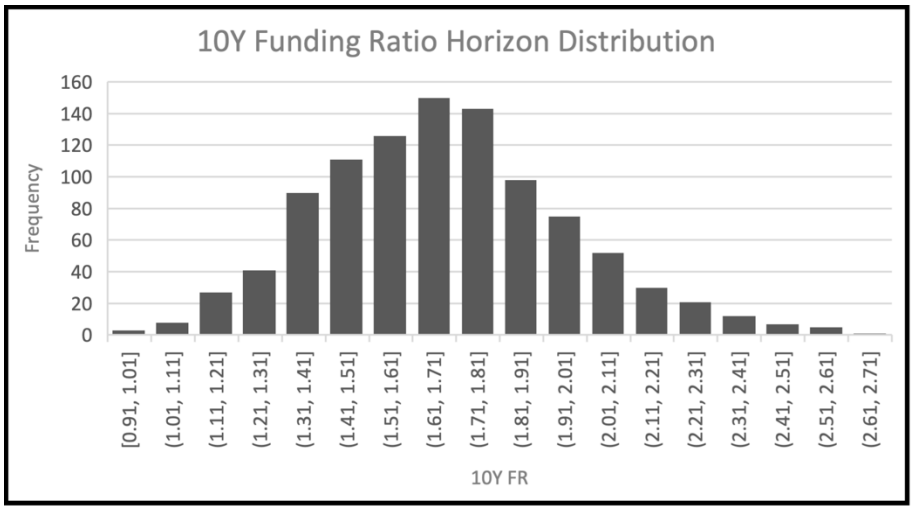

Figure 3: 10Y FR horizon distribution for 2025 mature asset allocation

MONTHLY PORTFOLIO STATISTICS					
Beginning Plan Assets (£m):	220.8	***FR Mean Change:***	**0.2384**	***FS Mean Change (£m):***	**53.6**
Beginning Plan Liabilities (£m):	176.6	***FR SD:***	**1.5610**	***FS 100P% VaR (£m):***	**-494.4**
Confidence Level: P =	0.95	***FR Skew:***	**-0.6590**	***FS 100P% CVaR (£m):***	**-784.3**
Beginning Funding Ratio:	**1.25**	***FR Kurtosis:***	**3.9926**	***FS Max Drawdown (£m):***	**-1573.2**

Figure 4: 2025 mature asset allocation monthly portfolio statistics

This paradoxically suggests less tail risk for 2011 allocations with greater equity weighting than for 2025 with the majority weighting in bonds. Indeed, the 2011 asset portfolio is more diversified across major asset classes than the 2025 portfolio and greater average FRs in 2025 than in 2011 may invite more proportional risk exposure, all else equal, but the fact that the standard deviation of the 2025 portfolio is lower than in 2011 (1.6 < 1.9) mechanically produces greater kurtosis.

Mean changes in FRs are positive across all plan allocations and liability types suggesting no structural problems in long-term funding for DB plans, assuming the 2013-2025 returns regime the simulator draws from is reflected in future asset class returns.

Maximum plan surplus/deficit drawdowns for all plan allocations over the 12-year period occur in March 2020, around the start of the COVID pandemic lockdowns.

For a typical overmature 2025 plan, FS/D monthly 95% VaR and CVaR per £1m of liabilities increase roughly linearly with FR, with 95% VaR exceeding liabilities for FR > 0.4 and 95% CVaR exceeding liabilities for FR > 0.25 (See Figure 5). Hence short-term expected losses are acute given current typical asset allocations.

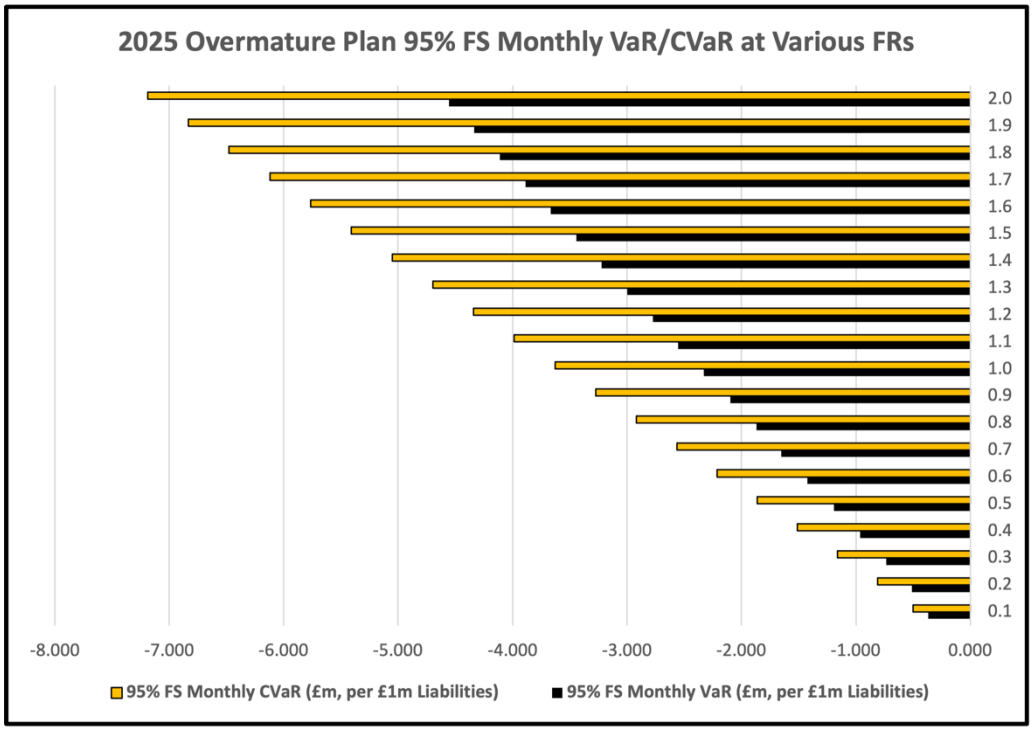

Figure 5: 2025 overmature plan 95% funding surplus/deficit monthly VaR/CVaR at various funding ratios

Comparing the lowest values in the PPF Purple Book 2011 and 2025 risk exposures (relative to an asset base of 100) reveals greater combined equity and interest rate downside risk for the typical 2025 plan assets compared to 2011 (see Figures 6 and 7). However, given that the 2025 allocation allocates more to bonds, the offsetting bond-like liability movements may offset interest rate risk for the plan surplus.

Table 5.2 | Impact of changes in gilt yields and equity prices on assets from a base of 100, 31 March 2011

Assets relative to a base of 100							
Movement in equity prices	Movement in gilts yields						
	-0.3pp	-0.2pp	-0.1pp	0.0pp	0.1pp	0.2pp	0.3pp
7.5%	104.4	104.0	103.6	103.2	102.8	102.4	102.0
5.0%	103.3	102.9	102.5	102.1	101.7	101.4	101.0
2.5%	102.3	101.9	101.5	101.1	100.7	100.3	99.9
0.0%	101.2	100.8	100.4	**100.0**	99.6	99.2	98.8
-2.5%	100.1	99.7	99.3	98.9	98.5	98.1	97.8
-5.0%	99.1	98.7	98.3	97.9	97.5	97.1	96.7
-7.5%	98.0	97.6	97.2	96.8	96.4	96.0	95.6

Source: PPF / The Pensions Regulator

Figure 6: PPF Purple Book 2011 asset sensitivities to interest rates and equity movements

Figure 5.8 | Impact of changes in gilt yields and equity prices on assets from a base of 100 as at 31 March 2025

Small changes in gilt yields have a slightly larger impact on assets than small changes in equity prices.

Assets relative to a base of 100							
Movement in equity prices	Movement in gilt yields						
	-0.3pp	-0.2pp	-0.1pp	0.0pp	0.1pp	0.2pp	0.3pp
7.5%	105.1	103.9	102.6	101.4	100.2	99.0	97.8
5.0%	104.7	103.4	102.2	100.9	99.7	98.5	97.3
2.5%	104.2	102.9	101.7	100.5	99.2	98.1	96.9
0.0%	103.8	102.5	101.2	**100.0**	98.8	97.6	96.4
-2.5%	103.3	102.0	100.8	99.5	98.3	97.1	96.0
-5.0%	102.8	101.6	100.3	99.1	97.9	96.7	95.5
-7.5%	102.4	101.1	99.8	98.6	97.4	96.2	95.0

Source: PPF

A 2.5 per cent increase in equity prices would raise scheme assets by 0.5 per cent (shaded). A 0.3 percentage point decrease in gilt yields would increase scheme assets by 3.8 per cent (shaded).

Figure 7: PPF Purple Book 2025 asset sensitivities to interest rates and equity movements

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